

ASTR 302 Fall 2026

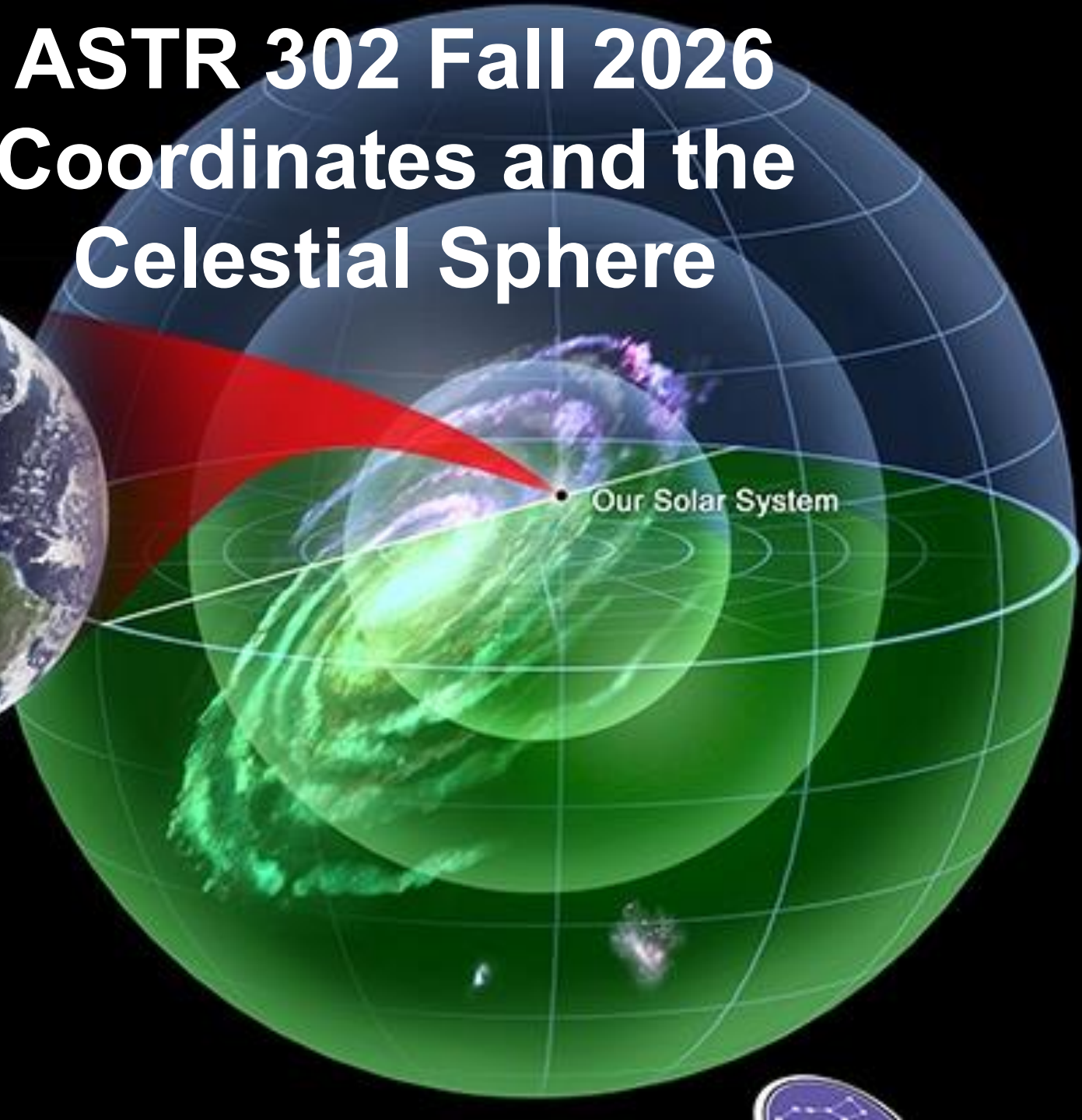
Coordinates and the Celestial Sphere



**Sloan Foundation
Telescope
New Mexico, U.S.A.**



**du Pont Telescope
Chile**



Agenda – September 2, 2026

- Class Logistics
- How Coordinates are Defined on the Celestial Sphere
- How and why certain astronomical objects move over time

Recommended Reading: Birney Chapter 1 and 4

Class Logistics

- Reach out to your point of contact/TAs for your final project to discuss feasibility.
- Group 4, you have 5 people now.
- Observing Proposals (**individual**) are Due Friday at 11:59 pm (**9/11**) on D2L

Looking Ahead

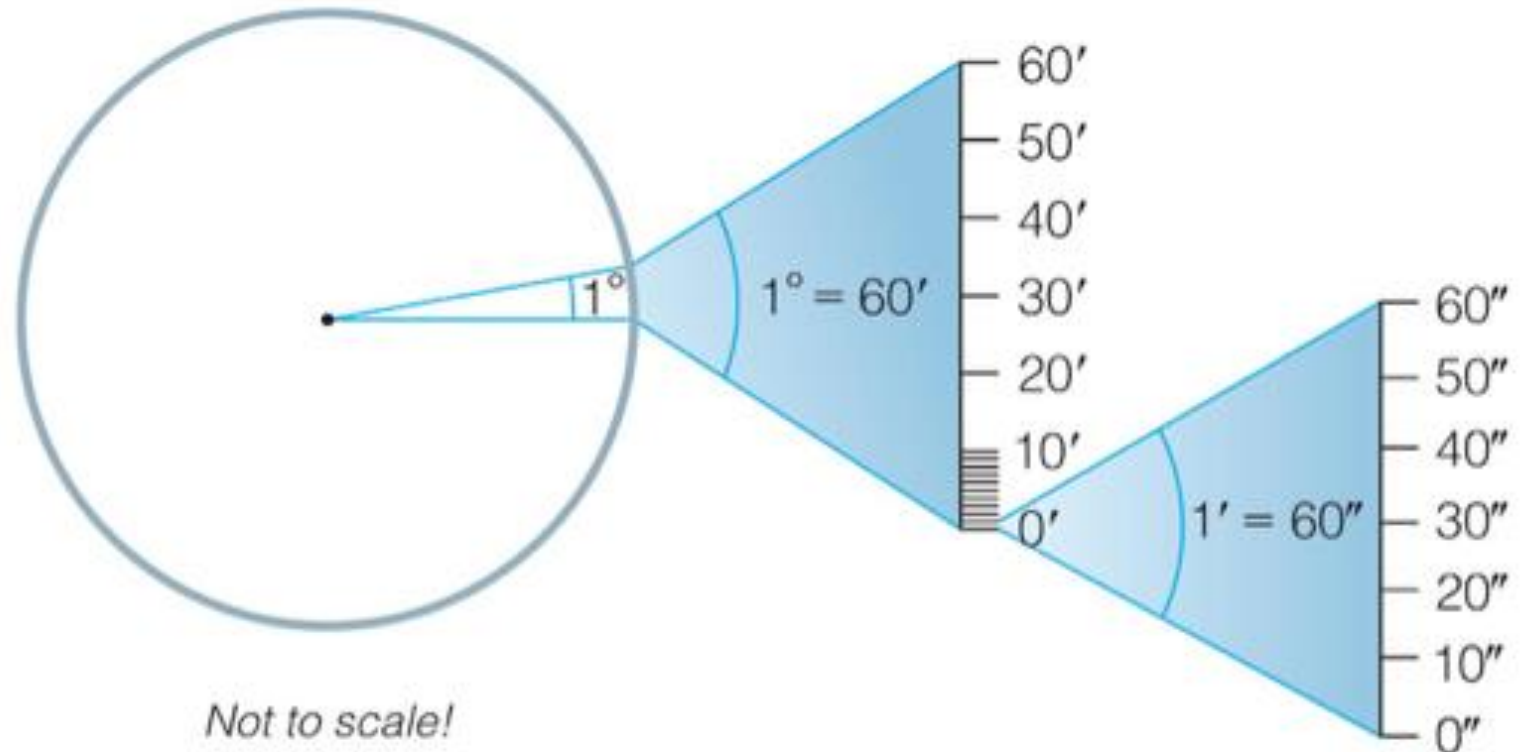
- 5 – minute presentations + questions describing your project (**Monday 9/21**)
(**5% of your Final Project Grade**)

Your TAs and Dr. Miles will grade you. Dr. Miles will moderate and ask ?s

We Describe Positions on the Sky in Angles

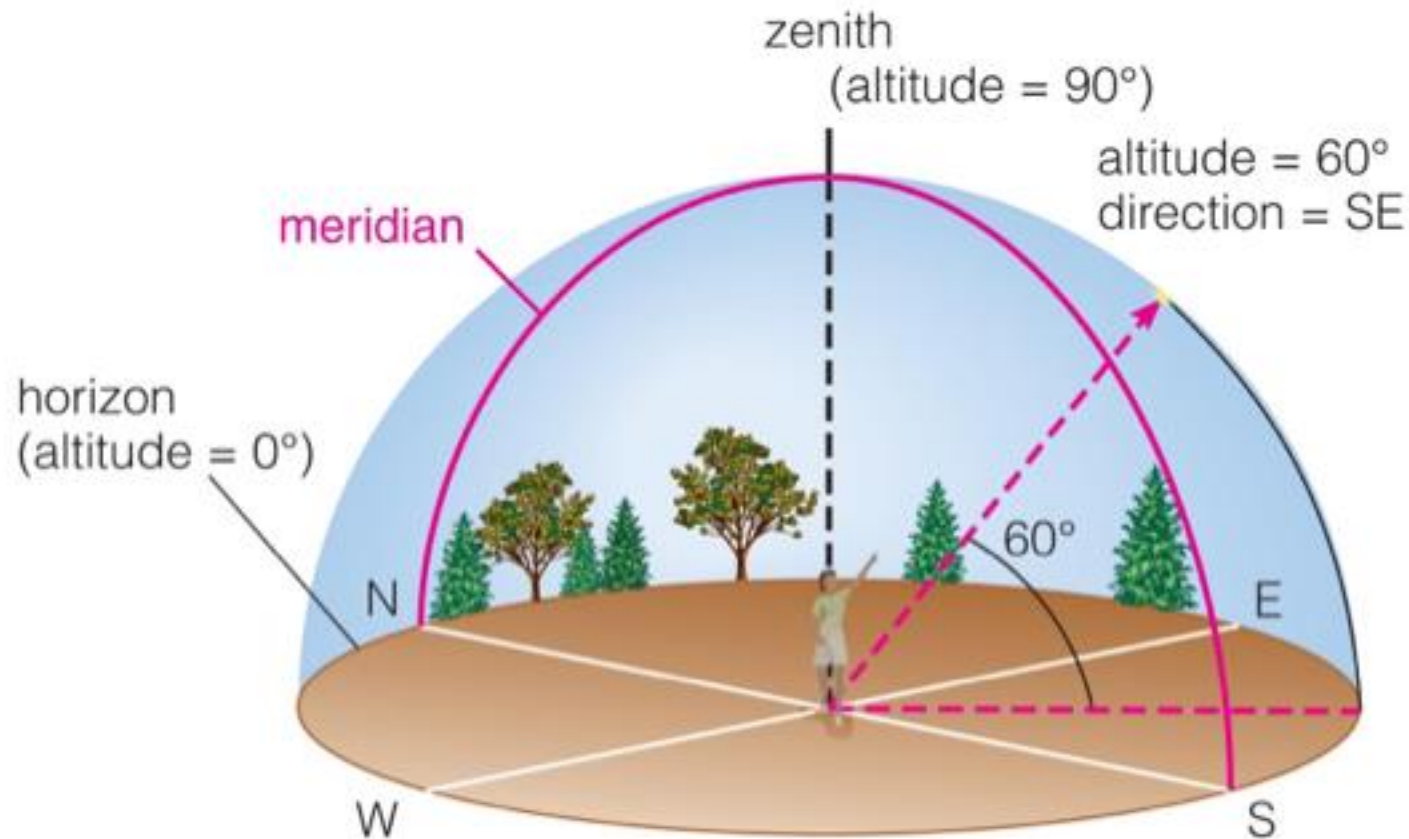
Angular Measurements

- Full circle = 360°
- $1^\circ = 60'$ (arcminutes)
- $1' = 60''$ (arcseconds)
- 1 hour = 15 degrees

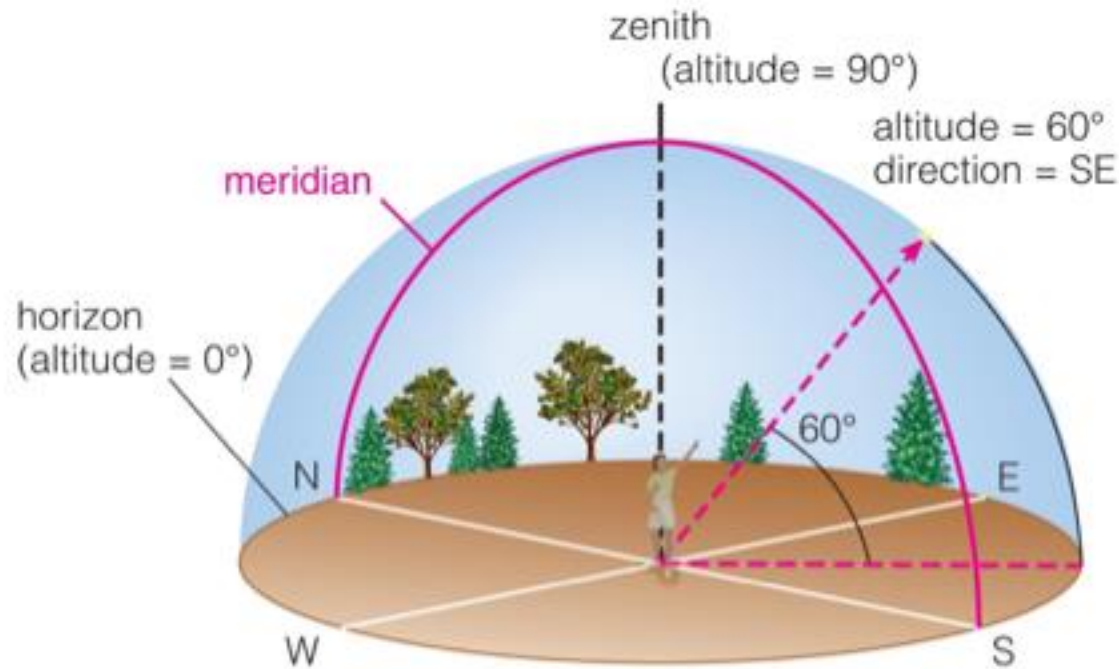


The Local Sky: Altitude - Azimuth System

An object's **altitude** (above horizon) and **azimuth** (direction along horizon) specifies its location in your local sky



The Local Sky: Altitude - Azimuth System

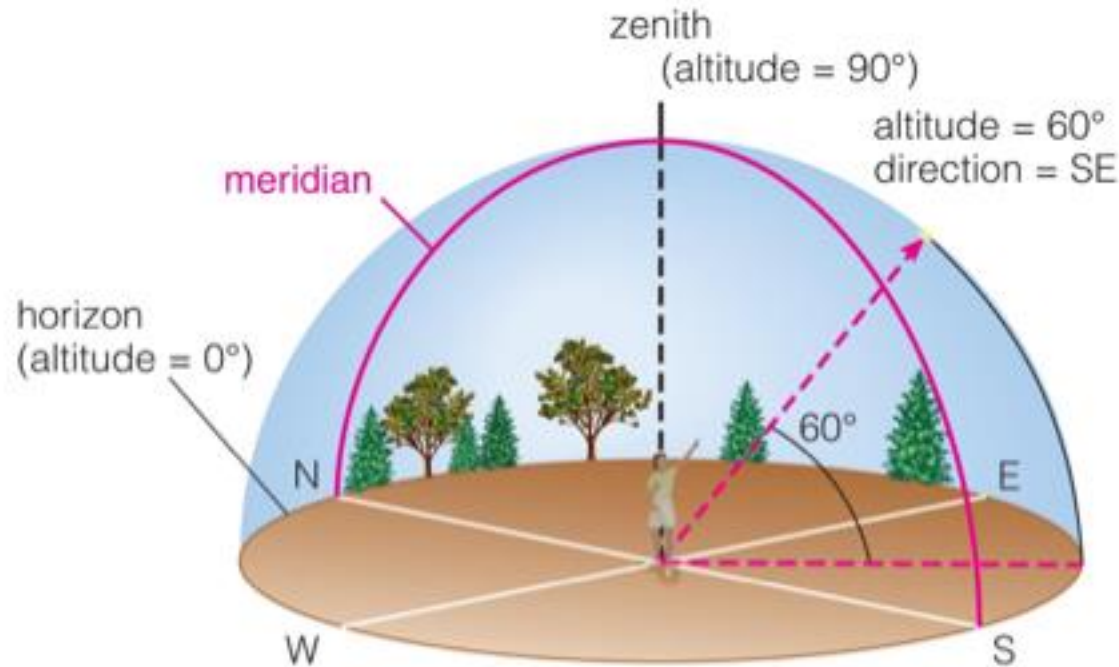


Zenith: The point directly overhead

Horizon: All points 90° away from zenith

Meridian: Line passing through zenith and connecting N and S points on horizon

The Local Sky: Altitude - Azimuth System

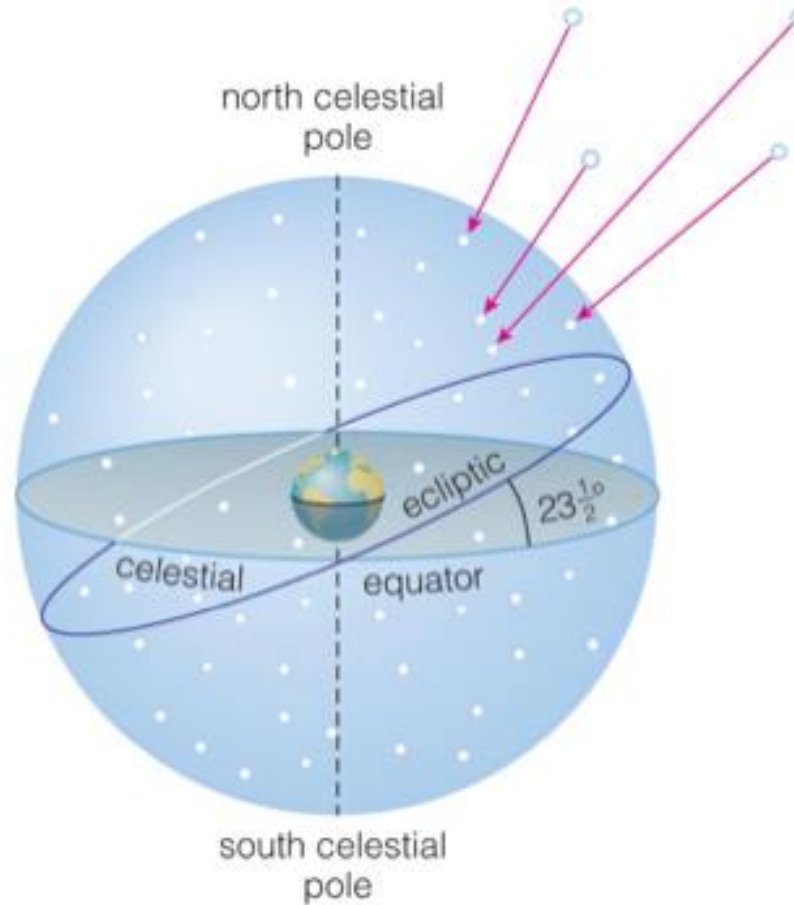


The alt-az system is very intuitive, but making the horizon your reference causes issues.

The coordinates are not “fixed” because Earth rotates towards the East.

In the alt-az system coordinates change over time.

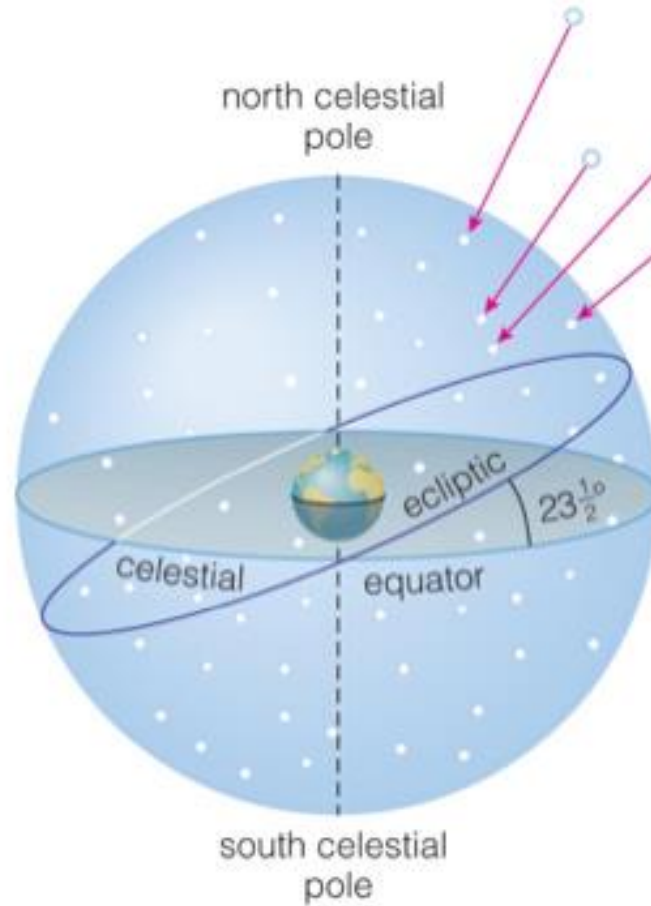
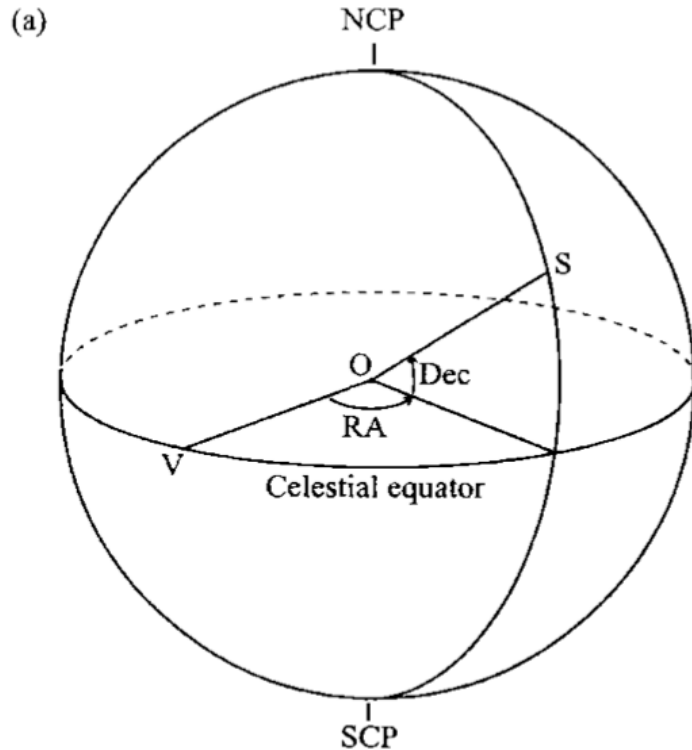
The Celestial Sphere and Equatorial Coordinates



Stars at different distances all appear to lie on the celestial sphere.

Ecliptic is Sun's apparent path through the celestial sphere.

The Celestial Sphere and Equatorial Coordinates

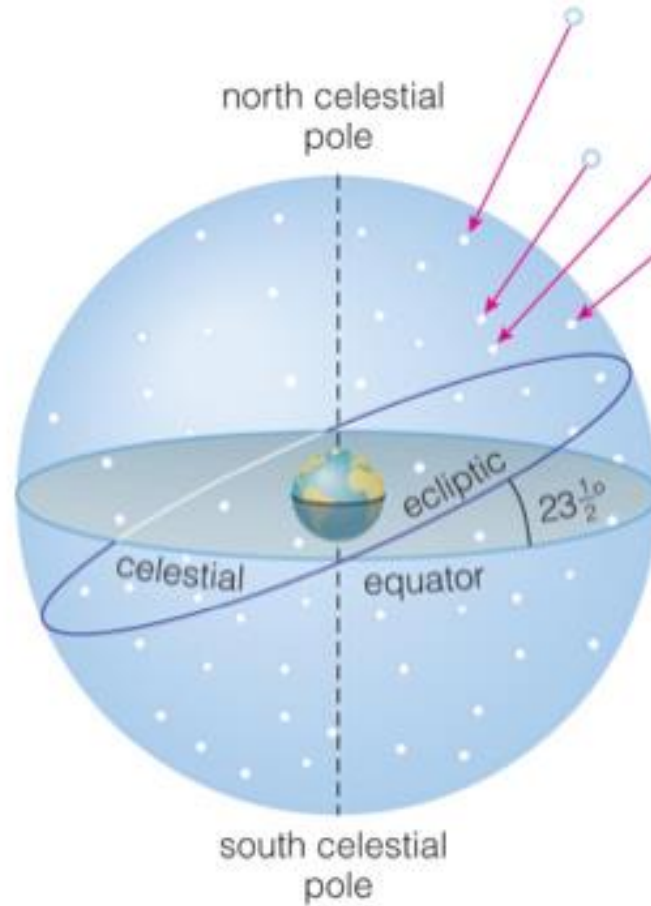
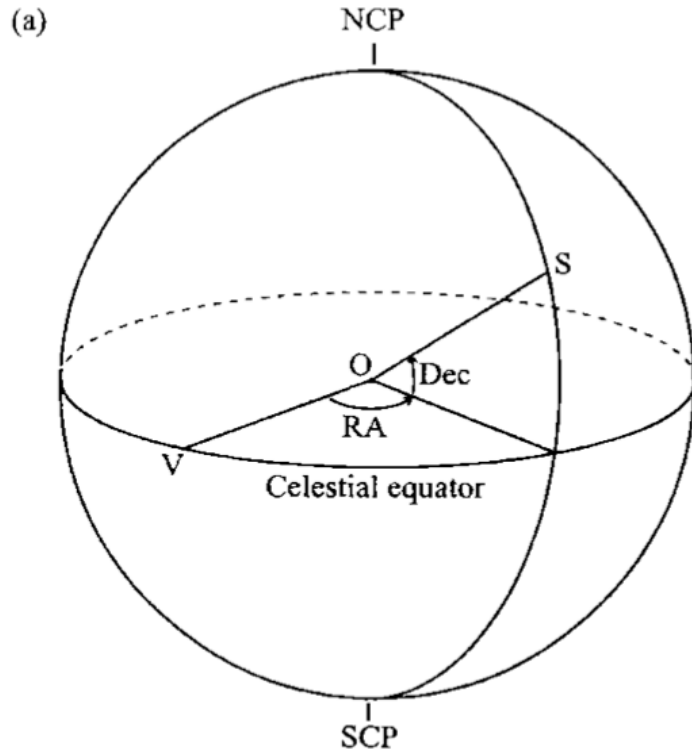


In the equatorial coordinate system, you quote two angles: **Right Ascension (RA)** and **Declination (DEC)**

The relative right ascension value is dependent on where the ecliptic crosses the celestial equator.

RA and **DEC** values are typically quoted with an **equinox year** and **epoch year**.

The Celestial Sphere and Equatorial Coordinates



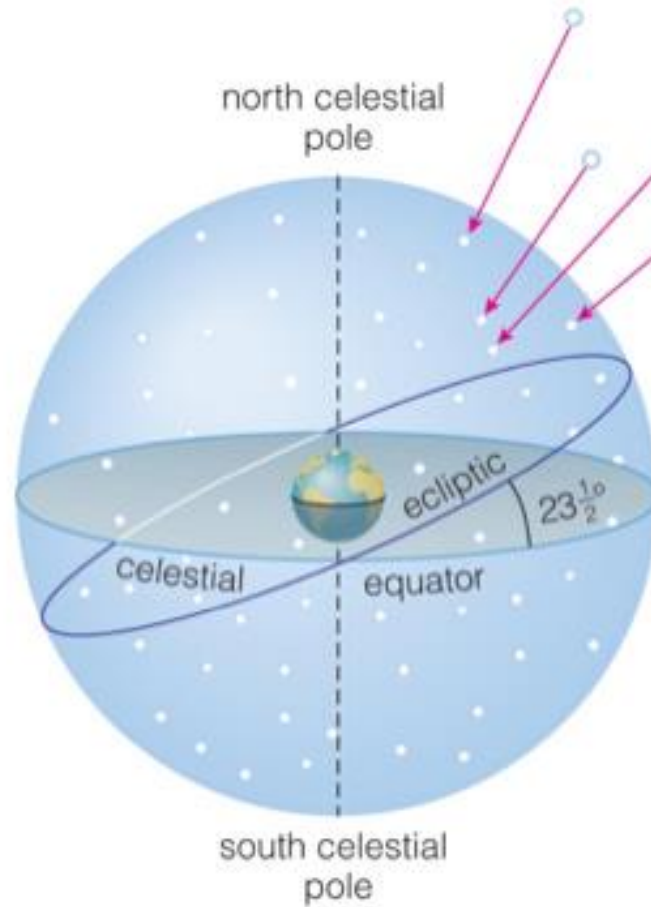
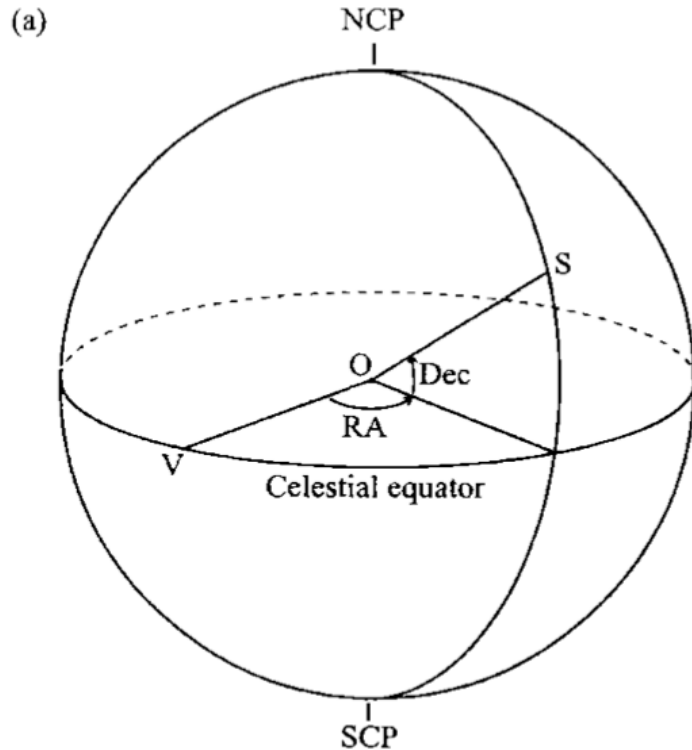
In the equatorial coordinate system, you quote two angles:
Right Ascension (RA) and
Declination (DEC)

Format(s)

**RA – hour: minute: seconds
(hh:mm:ss)**

**DEC – degrees: minutes: seconds
(dd:mm:ss)**

The Celestial Sphere and Equatorial Coordinates



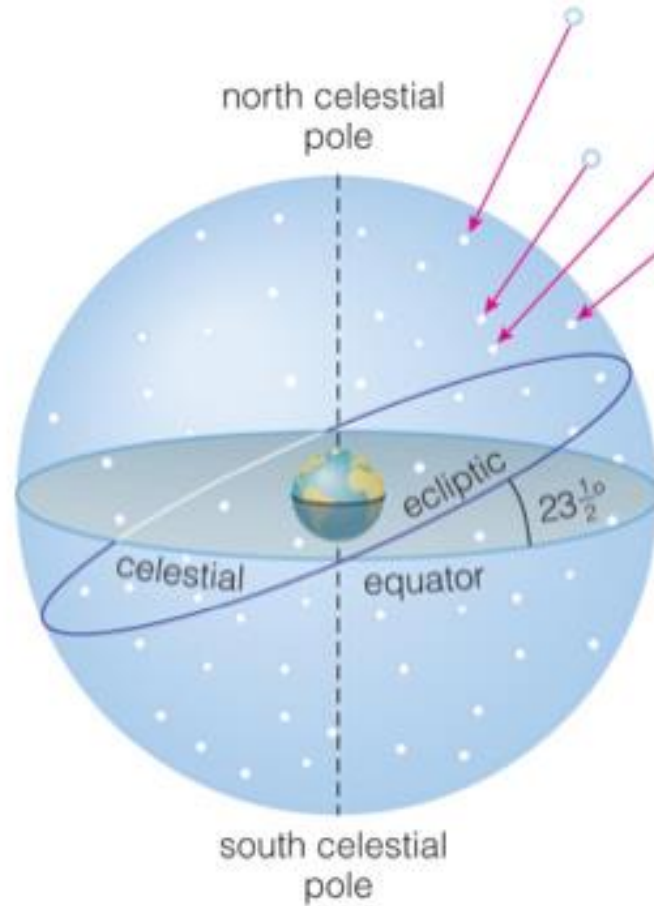
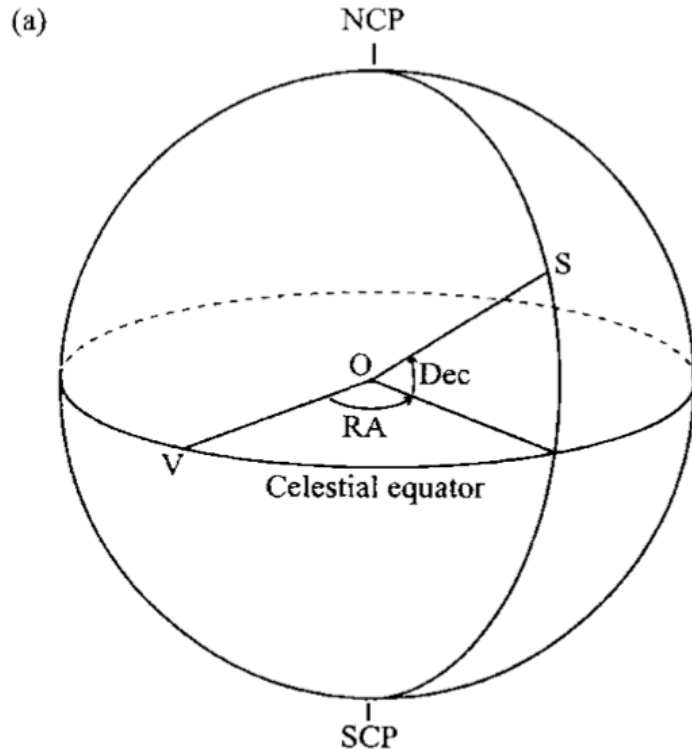
In the equatorial coordinate system, you quote two angles:
Right Ascension (RA) and
Declination (DEC)

Format(s)

RA – degrees (0, 360)

DEC – degrees (-90, 90)

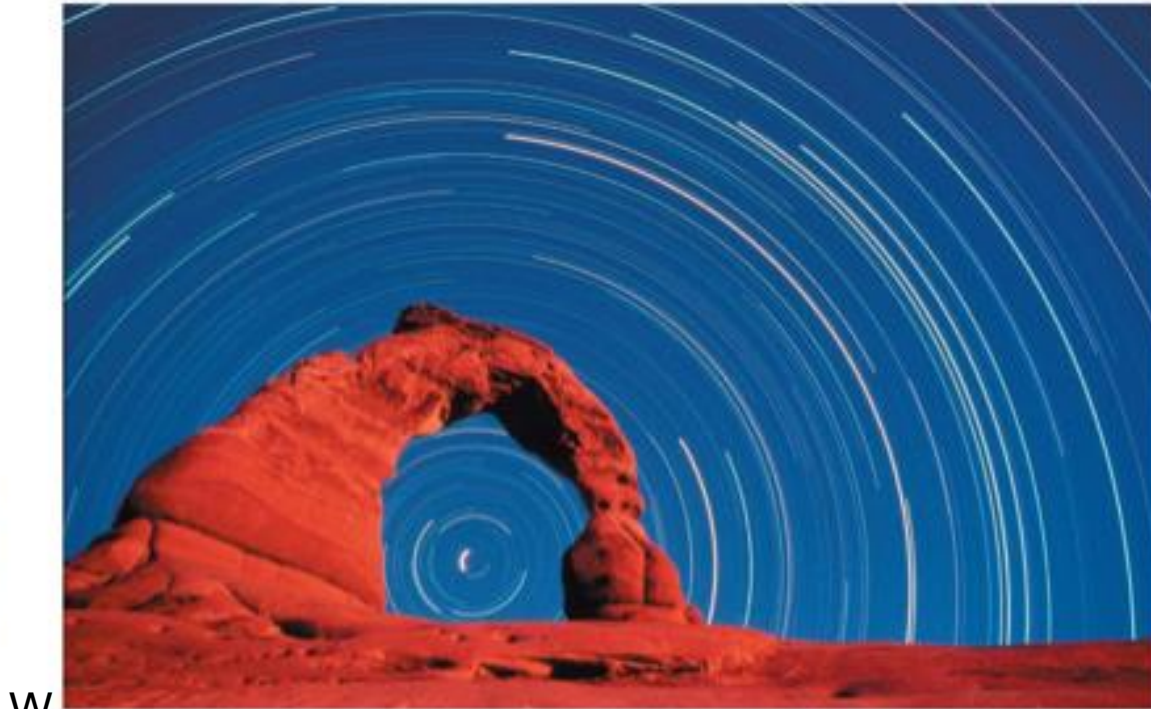
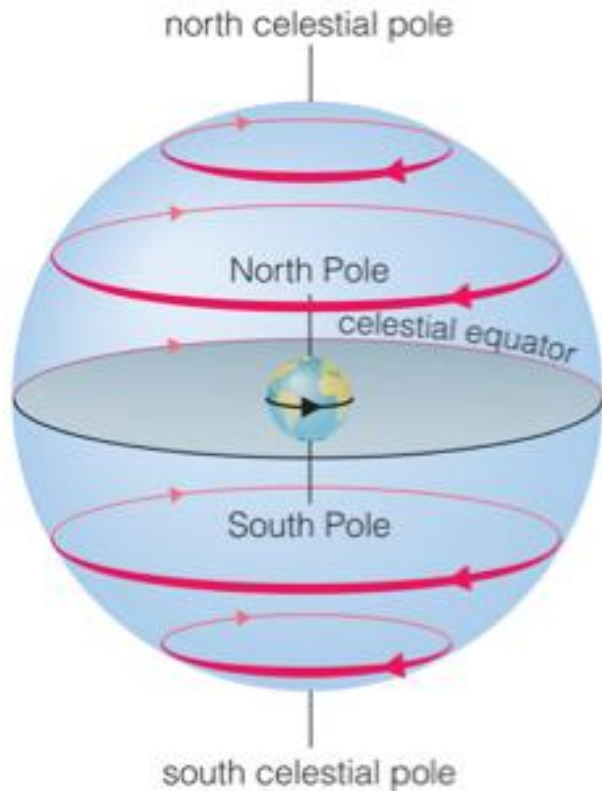
The Celestial Sphere and Equatorial Coordinates



Because the equatorial coordinate system is defined at a specific time it is “fixed” in RA and DEC.

Unlike the Alt-Az system where a target's angles constantly change over time.

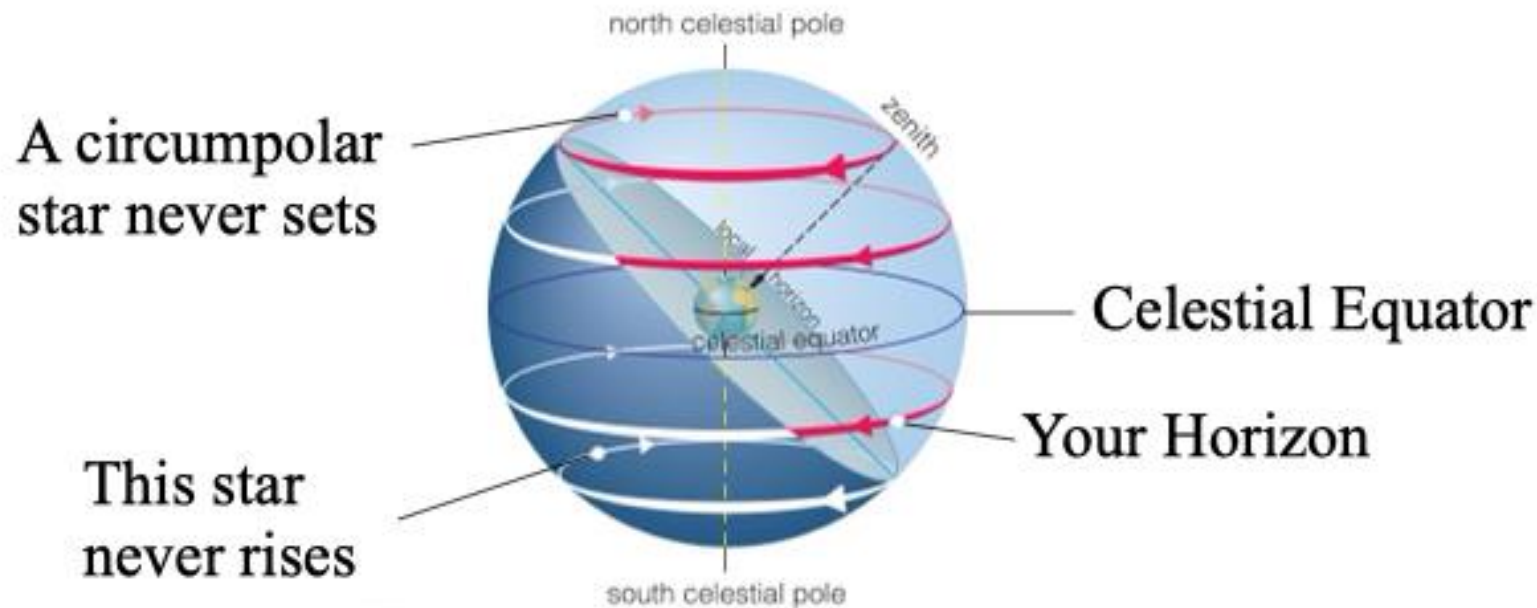
Stars Rise and Set in a Circle



Earth rotates west to east, so stars appear to circle from east to west.

Our view from Earth:

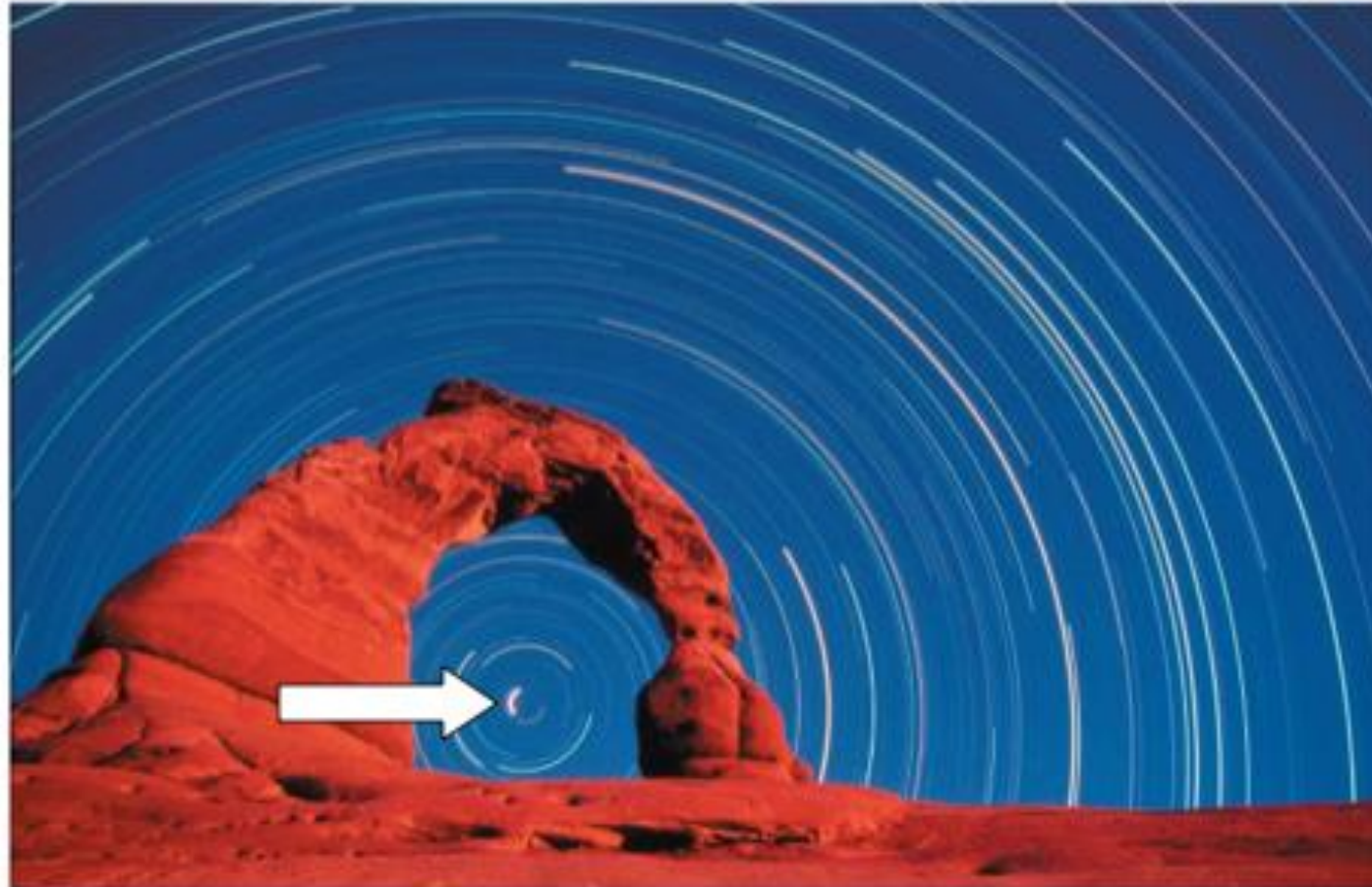
- Stars near the north celestial pole are circumpolar and never set.
- We cannot see stars near the south celestial pole.
- All other stars (and Sun, Moon, planets) rise in east and set in west.



Concept Check

What is the arrow pointing to?

- A. the zenith
- B. the north celestial pole
- C. the celestial equator

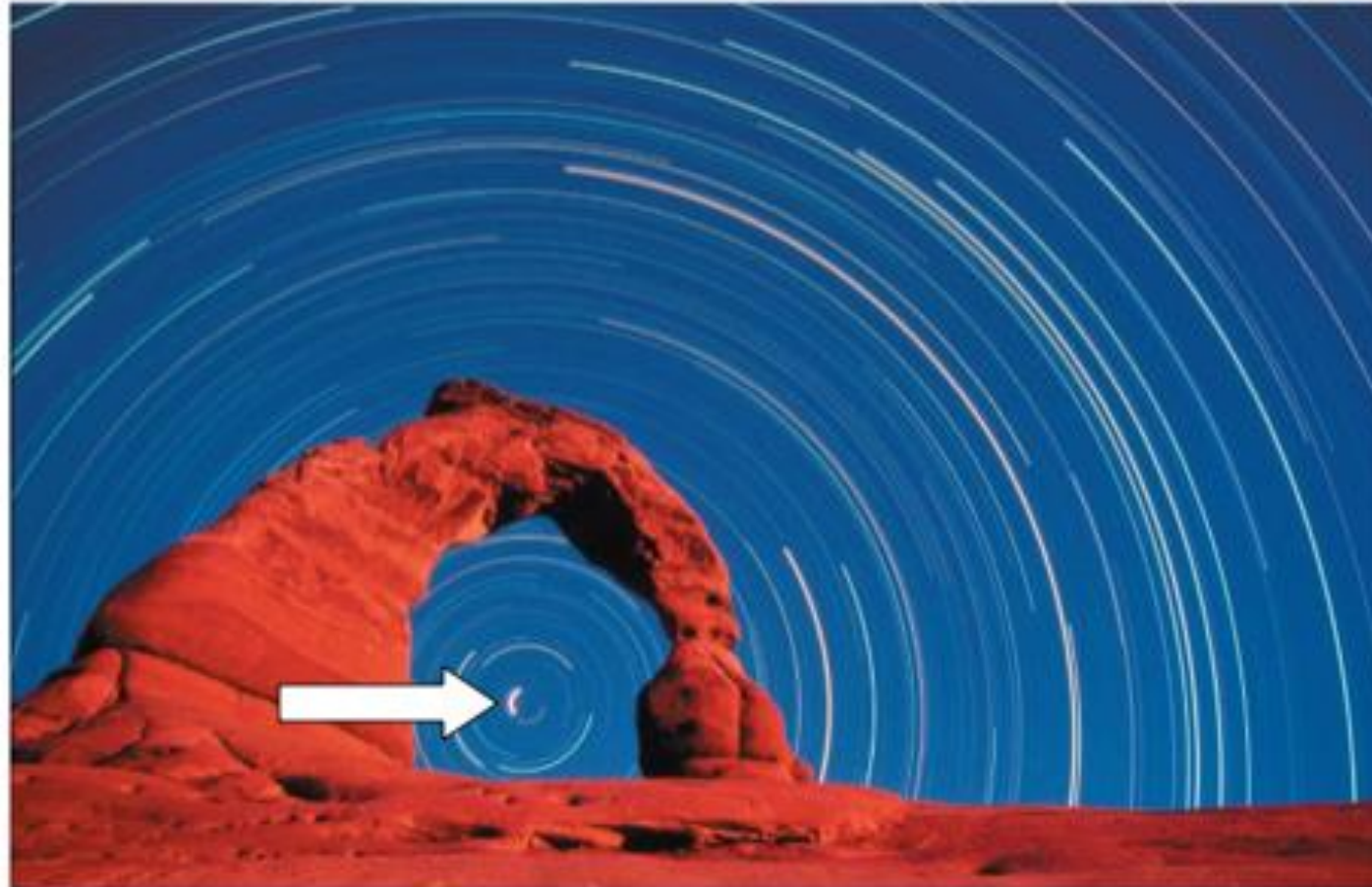


Concept Check

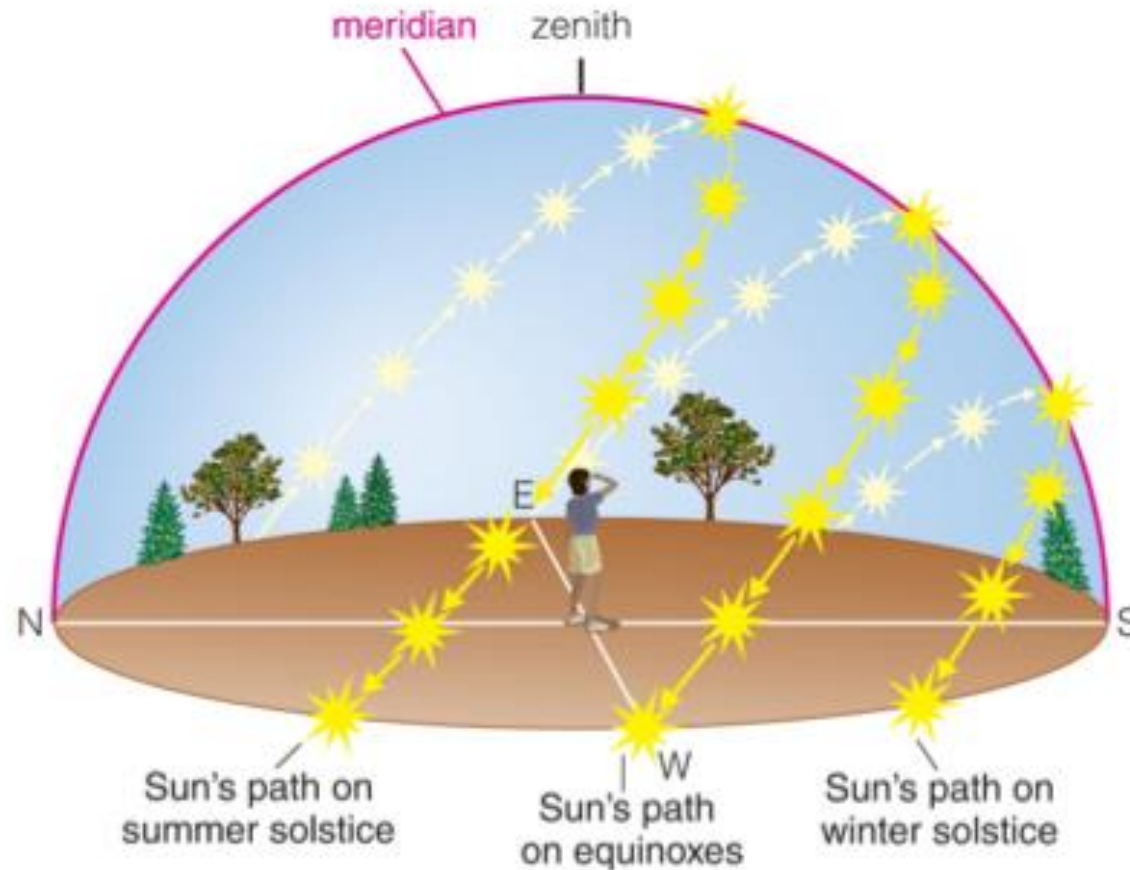
What is the arrow pointing to?

- A. the zenith
- B. the north celestial pole
- C. the celestial equator

B. Your latitude determines the height of the north or south celestial pole



Seasonal/Temporal Changes: The Sun

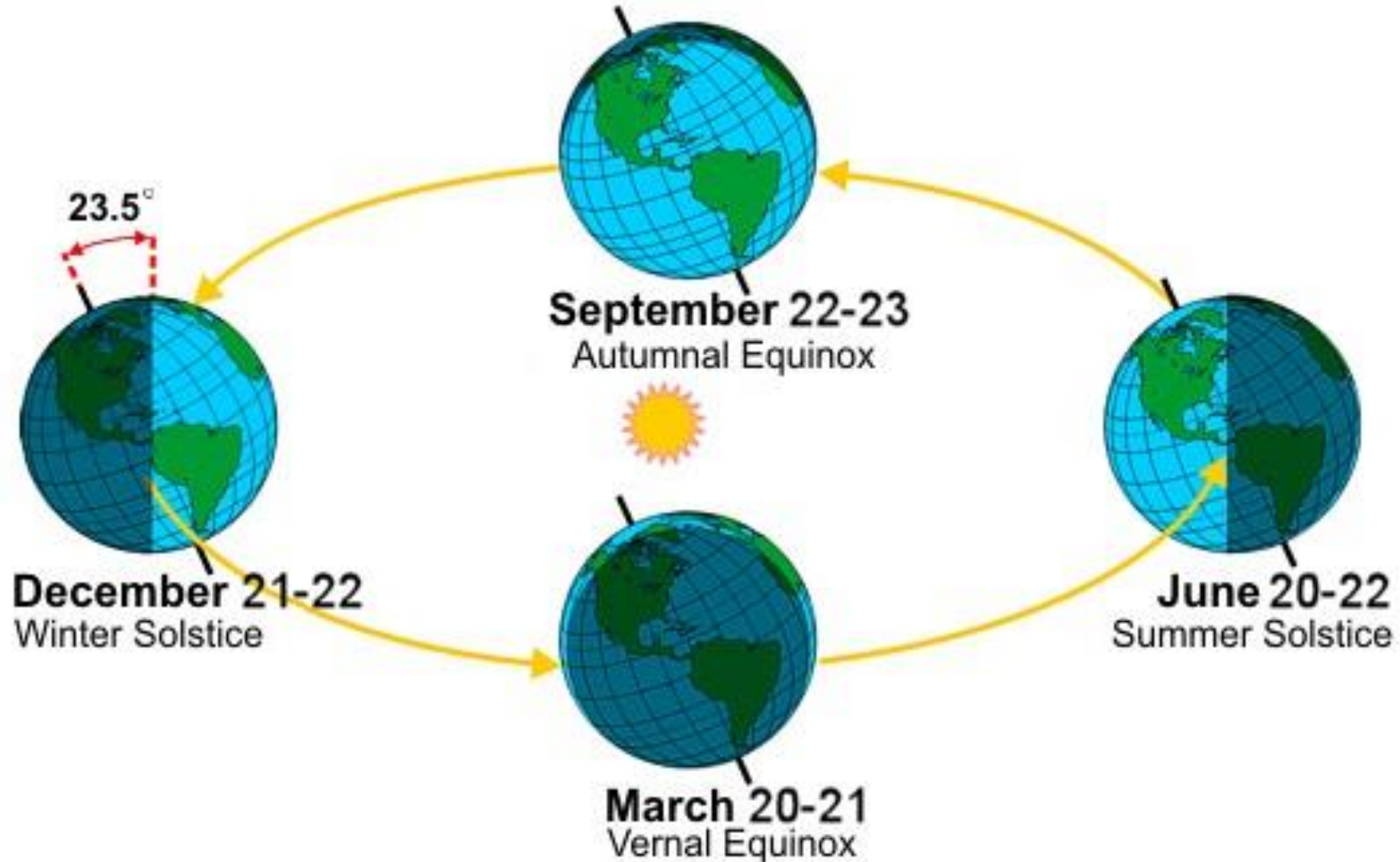


Summer solstice: Highest path, rise and set at most extreme north of due east.

Winter solstice: Lowest path, rise and set at most extreme south of due east.

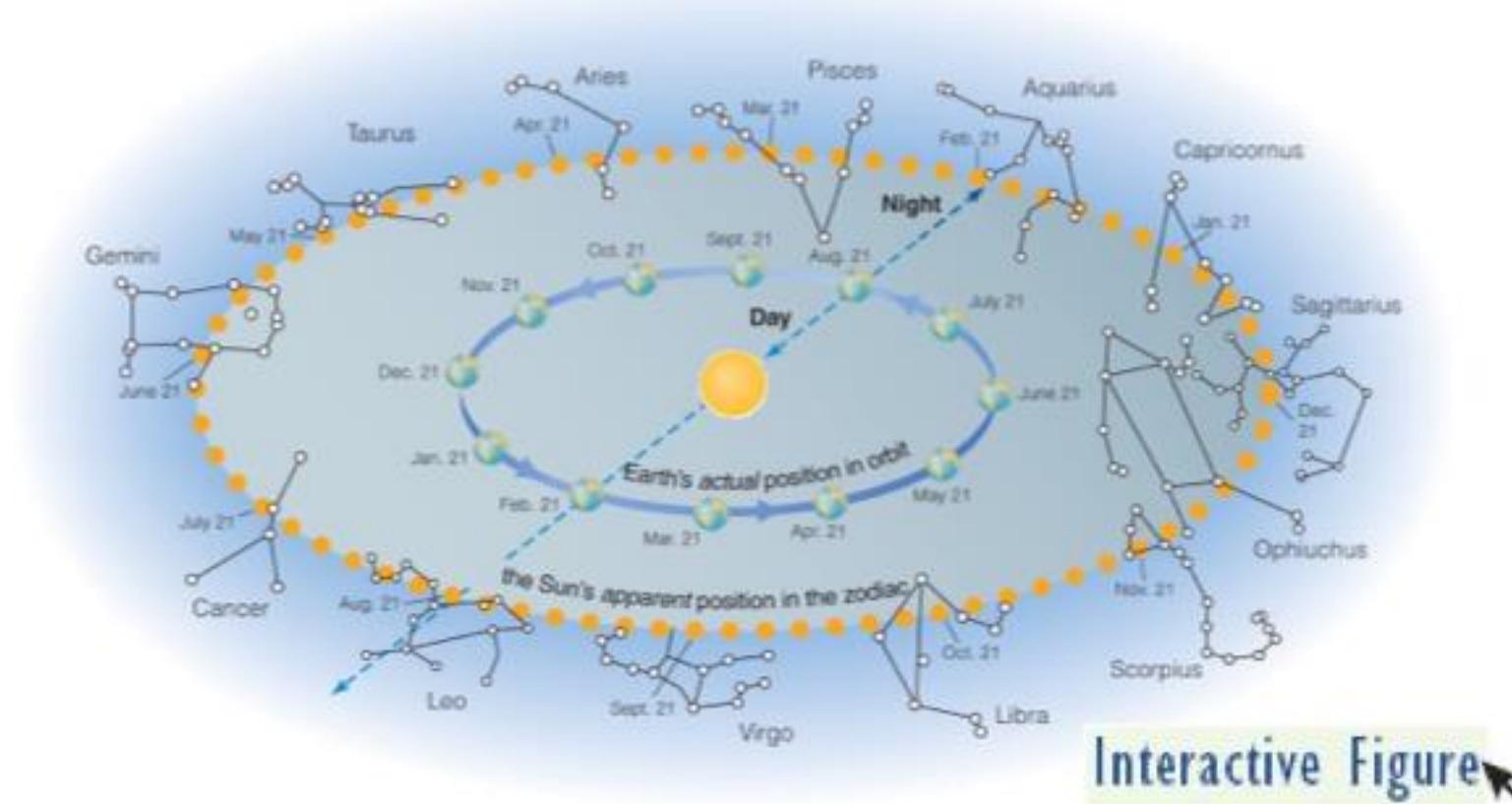
Equinoxes: Sun rises precisely due east and sets precisely due west.

Seasonal / Temporal Changes: The Sun



Seasonal / Temporal Changes: Constellations

- As the Earth orbits the Sun, the Sun appears to move eastward along the ecliptic.
- At midnight, the stars on our meridian are opposite the Sun in the sky.
- **The parts of the sky you can observe change with the seasons.**



The Milky Way Galaxy

